A Scalable Unsupervised Feature Selection With Orthogonal Graph Representation for Hyperspectral Images

Gulsen Taskin, *Senior Member, IEEE*, E. Fatih Yetkin, and Gustau Camps-Valls, *Fellow, IEEE*

***Abstract*— Feature selection (FS) is essential in various fields of science and engineering, from remote sensing to computer vision. Reducing data dimensionality by removing redundant features and selecting the most informative ones improves machine learning algorithms' performance, especially in supervised classification tasks, while lowering storage needs. Graph-embedding (GE) techniques have recently been found efficient for FS since they preserve the geometric structure of the original feature space while embedding data into a low-dimensional subspace. However, the main drawback is the high computational cost of solving an eigenvalue decomposition problem, especially for large-scale problems. This article addresses this issue by combining the GE framework and representation theory for a novel FS method. Inspired by the high-dimensional model representation (HDMR), the feature transformation is assumed to be a linear combination of a set of univariate orthogonal functions carried out in the GE framework. As a result, an explicit embedding function is created, which can be utilized to embed out-of-samples into low-dimensional space and provide a feature relevance score. The significant contribution of the proposed method is to divide an n-dimensional generalized eigenvalue problem into n small-sized eigenvalue problems. With this property, the computational complexity (CC) of the GE is significantly reduced, resulting in a scalable FS method, which could be easily parallelized too. The performance of the proposed method is compared favorably to its counterparts in high-dimensional hyperspectral image (HSI) processing in terms of classification accuracy, feature stability, and computational time.**

***Index Terms*— Dimensionality reduction, feature selection (FS), global sensitivity analysis, graph embedding (GE), hyperspectral image (HSI) analysis.**

I. Introduction

RECENT developments of imaging spectrometer technology in remote sensing have opened new opportunities to monitor the Earth's surface. Hyperspectral sensors, in particular, provide detailed information about different ground materials as they acquire hundreds of reflectance on different wavelengths for each pixel with a high spectral resolution. Therefore, they allow greater differentiation between similar ground cover classes than typical multispectral images [1], [2]. The distinct characteristics of the hyperspectral images (HSI) make them highly effective in a wide range of remote sensing applications, such as classification [3], anomaly detection [4], unmixing [5], and change detection [6]. However, a fundamental constraint for efficient utilization of hyperspectral imagery in such applications is the high number of spectral bands, which causes an increase in computational time and difficulty of data storage, transmission, and processing [6]. Additionally, noise and high correlation between the adjacent spectral bands decrease the performance of learning algorithms, especially with the limited training samples. Even some spectral bands might not include discriminant information for a specific application. To deal with this, feature selection (FS) can be used as an effective method to reduce the dimensionality of HSI without a considerable loss of information [7], [8], [9].

The FS identifies the most relevant and informative features from existing features. Therefore, they effectively decrease dimensionality, speed up data processing, enhance the capability of data visualization and interpretability, and increase the generalization ability of learning algorithms [10], [11], [12]. The FS methods can be categorized into three groups based on the availability of ground-truth data: supervised, semi-supervised, and unsupervised. Supervised FS requires labeled data, with each sample assigned to a class or a real number, whereas semi-supervised FS requires labeled and unlabelled data. Unsupervised FS (UFS) approaches select features by taking into account the natural characteristics of the data without requiring any labeled samples. As for supervised methods, collecting the labeled data is a time-consuming and expensive process in HSI analysis, resulting in limited ground-truth samples. Therefore, the UFS methods have some advantages over supervised approaches for HSI analysis: 1) they are unbiased and achieve better performance when prior information is unavailable; 2) they can avoid the risk of model overfitting; and 3) they are fast and efficient. Depending on the methodology used to select relevant features, either clustering or evaluation criteria, the UFS methods are grouped into three types: filter, wrapper, and embedding methods [13], [14]. Filter methods evaluate features based on criteria typically generated by analyzing data properties. Therefore, filter

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methods become more prominent than the other two in terms of their scalability for high-dimensional data analysis since the training of ML models is not involved [15], [16], [17]. In this work, we will investigate unsupervised filter-based FS in light of these considerations.

Recently, spectral (graph) learning has been effectively utilized as UFS methods in reducing the dimensionality of HSI [18], [19], [20], [21]. Such related methods perform well in selecting the relevant features as spectral learning aims to preserve the geometric structure of the original feature space while embedding data into a low-dimensional subspace. However, the main drawback of such methods is the high computational cost involved in solving an eigenvalue decomposition problem, which typically results in an inefficient model in terms of scalability [22], [23]. Several research papers address this as a significant issue and propose effective FS approaches, with the most relevant ones being covered in detail in Section II. However, none of these methods theoretically solves the scalability of the FS issue.

In this article, we address the scalability of the FS methods by combining the graph learning framework and representation theory. To deal with the scalability issue in the FS problem while still relying on solid graph theory, we developed an UFS method by imposing an orthogonality condition to each feature in the graph embedding (GE) framework via the high-dimensional model representation (HDMR) [24], [25], [26]. We solved the resulting optimization problem analytically, which advantageously divides the costly large-scale generalized eigenvalue problem into the much cheaper small-sized problems. HDMR is a method for representing a multivariate function as the sum of lower-order functions, including the constant term, univariate term, and interactions of multivariate terms. The HDMR is mainly used as a surrogate model to replace a highly cost function with a simpler one or global sensitivity analysis, measuring the uncertainty of the inputs on the model outputs. The use of the HDMR is first conducted by Kaya et al. [27] and Taşkın et al. [28] as a semi-supervised FS method where the effects of each feature are determined based on the sensitivity scores of the HDMR functions that are calculated using Monte Carlo integration schemes. Later, the HDMR adapted to a regression problem in the least-squares sense to solve the out-of-sample problem of the manifold learning [29]. The developed method is used by Erten et al. [30] for global sensitivity analysis for selecting polarimetric synthetic aperture radar (PolSAR) observables for crop biophysical variable estimation. Recently, for the first time, we cast the HDMR in the GE framework for feature extraction and FS problems [31], [32]. The main contributions of this study can be listed as follows:

1) imposing orthogonality condition as a constraint into the proposed GE framework, resulting in an FS algorithm;
2) partitioning a large-scale generalized eigenvalue problem into a set of small-sized eigenvalue problems;
3) designing a novel GE approach which is fully convenient for parallel processing;
4) proposing a mathematically tractable and computationally efficient eigenvalue-based FS mechanism.

The organization of this article is as follows. First, in Section II, the related literature is presented. In Section III, the fundamentals of the HDMR are briefly given. In Section IV, the proposed method is presented in detail. Sections V and VI present and discuss the results of the proposed method's performance compared to state-of-the-art FS methods. Finally, Section VII concludes the article by highlighting the methods' strengths.

II. Related Work

In unsupervised learning, the filter approaches seek to estimate and optimize different criteria when selecting features, including statistical criteria (like correlation or covariance), information-theoretic criteria (such as entropy and mutual information), spectral decompositions (via manifold learning), and sparsity of the signal representation. Among them, spectral and sparse learning approaches have proven to be more effective in learning the relevant features as they efficiently characterize the geometric relations between the samples using graph representations [33], [34], [35], [36], [37]. Spectral learning methods attempt to uncover a true low-dimensional structure using eigenvalue analysis. At the same time, sparse representations reduce dimensionality using a sparsity constraint to ensure a trade-off between performance and simple representations. One of the well-known methods utilizing spectral learning is Laplacian score (LapScore), introduced by He et al. [38], which was based on the standard GE method called Laplacian Eigenmaps [39]. The LapScore selects the relevant features based on a local relationship between high- and low-dimensional space constructed with a neighborhood graph. Spectral FS (SPEC) [40], an extension of LapScore, evaluates the feature relevancy concerning its consistency with graph Laplacian. These two methods consider mapping from high-dimensional to low-dimensional space as linear. Even though such graph-based methods effectively select the relevant features, they pose a high computational cost for solving a large-scale generalized eigenvalue problem [41], [42]. Several attempts have been made to improve the efficiency and scalability of FS algorithms for large-scale data analysis [43], [44]. For example, Sun et al. [45] cast the GE problem as a least-squares regression optimization problem to reduce the cost of solving a generalized eigenvalue problem. Benabdeslem and Hindawi [46] propose a fast semi-supervised FS method providing a new score function to evaluate the relevance of features based on both the local structure of the unlabelled data distribution and pairwise constraints utilizing the class-wise information of labeled data. Since employing all features for data structure discovery in such graph-based methods may result in noise amplification, Chen et al. [41] developed a unified FS framework in which a binary feature selector is introduced to naturally handle the feature cardinality in the least-squares formulation with a sparse representation.

Sparse learning in conjunction with spectral learning is another methodology that aims to reduce the FS method's complexity and speed up the algorithm [47], [48], [49]. One of the earliest methods in this category is multicluster FS (MCFS) [50]. The method detects the clusters using sparse learning. It determines the importance of each feature with

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respect to the highest absolute values of the coefficients of a regression model with a ℓ_1-norm regularization. Due to the efficiency of $\ell_{2,1}$-norm minimization, unsupervised discriminative FS (UDFS) is proposed by Yang et al. [51] that incorporates discriminant analysis into $\ell_{2,1}$-norm minimization. Similarly, Zhang et al. [52] proposed a clustering-based FS method that uses $\ell_{2,1}$-norm regularization to make the solution sparser and hence more robust to noisy features. It should be noted that the $\ell_{2,1}$-norm-based methods involve iterative matrix inverse operations in the optimization process, resulting in a high computational cost, which makes them infeasible for large-scale data analysis in terms of the number of features. To overcome this, Yan et al. [53] explicitly impose non-negativity and the $\ell_{2,1}$-norm constraints onto the objective function to obtain a feature weights matrix that is row-sparse and non-negative. Similarly, Shang et al. [33] proposed a non-negative spectral learning and sparse regression-based dual-graph regularized FS method. They generate iterative and alternative updating rules to optimize the objective function, enabling it to select the representative features quickly and efficiently. Even though $\ell_{2,1}$-norm constraint delivers satisfactory results, $\ell_{2,0}$-norm is more desirable [54]. As finding an appropriate regularization parameter to select a specific number of features with $\ell_{2,1}$-norm regularization takes a long time, Pang et al. [55] proposed a novel method to solve the $\ell_{2,0}$-norm constrained problem for FS. In sparse learning-based methods, the optimization problem must be solved iteratively due to the non-convex structure of the sparsity constraint. Therefore, it is worth noting that the performance of such methods strongly depends on an efficient design of the iterative optimization scheme, and convergence may not always be guaranteed.

III. High-Dimensional Model Representation

HDMR is a serial expansion where each term is represented as a linear combination of orthogonal monomial functions: univariate, bivariate, etc. A d-dimensional function can be represented as follows:

$$f(x_1,\ldots,x_d) = f_0 + \sum_{i=1}^{d} f_i(x_i) + \sum_{1\le i<j\le d} f_{ij}(x_i,x_j) + \cdots + f_{12\ldots d}(x_1,\ldots,x_d) \quad (1)$$

where f_0 corresponds to a constant term representing the mean response of the model. The other additive terms on the right-hand side of (1) provide the effects of the variables independently $[f_i(x_i)]$ or jointly $[f_{ij}(x_i,x_j)]$ on the model output $f(x)$. The HDMR expansion can be truncated up to second-order terms for practical purposes, and it typically yields a good approximation for a specific function [56]. Each term satisfies the following orthogonality condition, which is a direct consequence of well-known Sobol's condition in domain $\mathbf{K}^d$ [57]:

$$\int_{\mathbf{K}^d} f_{i_1,i_2,\ldots,i_k} f_{j_1,j_2,\ldots,j_l} d\mathbf{x} = 0 \quad (2)$$

where $f_{i_1,i_2,\ldots,i_k}$ and $f_{j_1,j_2,\ldots,j_l}$ correspond to the right-hand side terms of (1). Due to this orthogonality constraint, one can achieve the HDMR terms by computing multidimensional integrals. A numerical integration scheme, such as Monte Carlo, can be employed when the analytical expression of the function is not known or too complex. However, this may result in a huge computational cost as a 2^d, the number of model evaluations of random samplings required for computing HDMR terms. To reduce the cost as well as sampling effort for Monte Carlo integration, random sampling HDMR (RS-HDMR) is suggested [56], which is a method for representing each term as a linear combination of orthogonal polynomial basis

$$f_i(x_i) \approx \sum_{r=1}^{p} \alpha_r^i \phi_r(x_i) \quad (3)$$

where p corresponds to the order of the polynomial and α_r^i values are the coefficients of the polynomials, $\phi_r(x_i)$.

The RS-HDMR requires many random samplings to cover the entire feature space. The random samplings are typically created using space-filling methods such as Latin hypercube sampling [58], Sobol's sequence [59], uniform designs [60], and Hammersley's sequence [61]. For the relevant sample, the function (f) evaluation or a black-box evaluation, if one is available, is required to compute model output. The RS-HDMR, on the other hand, is inappropriate for supervised learning since the samples and their associated model outputs are already given as a training set and do not follow an ordered sampling scheme. In such cases, the RS-HDMR terms can be computed by designing a specific loss function, such as the ℓ_2-norm empirical function, and then solving a least-squares problem to obtain the coefficients [56]. In an unsupervised learning setting, however, there is no model output, and the objective is to identify the underlying correlations in the data. In this study, we exploit the RS-HDMR in an unsupervised setting by formulating a GE framework to represent the coordinates of the latent space through RS-HDMR and imposing orthogonality constraints. Solving this optimization problem leads to relatively small-sized eigenvalue problems, from which the RS-HDMR terms can be calculated analytically. The coefficients of the RS-HDMR terms allow for direct computation of Sobol indices that might be used for feature ranking.

IV. Proposed HDMR Method for FS

Let us start by fixing the notation. Let $\mathbf{X} = \{x_1,\ldots,x_d\} \in \mathbb{R}^{n\times d}$ be a data (or design) matrix, where d and n are the number of features and samples, respectively. Each $x_i \in \mathbb{R}^{n\times 1}$ corresponds to feature vectors. Linear dimensionality reduction finds an $\mathbf{A} \in \mathbb{R}^{n\times r}$ projection matrix, where r denotes the dimension of subspace, to map high-dimensional data into a low-dimensional subspace with $y_i = \mathbf{A}x_i$ for each x_i and y_i vectors or $\mathbf{y} = \mathbf{AX}$ in compact matrix form. Locality preserving projection (LPP) [62] uses a similar approach, incorporating an adjacency matrix and a graph Laplacian.

In contrast to the linearity assumption, the HDMR may be used to achieve a nonlinear representation between the data matrix and its lower-dimensional representation. Assume that nonlinear mapping function to 1-D space (where $r = 1$)

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$y = f(\mathbf{X})$ is defined as follows:

$$y = f_1(x_1) + f_2(x_2) + \cdots + f_d(x_d) \tag{4}$$

where each x_i corresponds to a feature vector of associated $i-$th sample, whereas $f(x_i)$ refers to ith HDMR term. Note that f_0 vanishes when the data matrix is centered. Each HDMR term can be written as the following linear combination of p Legendre polynomials:

$$f_i(x_i) = \sum_{j=1}^{p} \alpha_j^i \phi_j^i(x_i). \tag{5}$$

Now, by substituting (5) into (4), the following compact matrix form is obtained:

$$y = \Phi_1\psi_1 + \Phi_2\psi_2 + \cdots + \Phi_d\psi_d. \tag{6}$$

Each HDMR term in (6) is expressed as an matrix–vector product of the matrix $\Phi_\mathbf{i}$ and vector ψ_i, as shown

$$\Phi_i = \begin{bmatrix}\phi_1^i & \dots & \phi_p^i\end{bmatrix}$$
$$\psi_\mathbf{i} = \begin{bmatrix}\alpha_1{}^i & \alpha_2{}^i & \dots & \alpha_p{}^i\end{bmatrix}^T.$$

Note that ϕ_j^i is a vector, obtained by applying the jth Legendre polynomial to the feature vector x_i, and α_j^i is an unknown scalar, corresponding to Legendre's coefficient. Hence, we can define the nonlinear transformation to 1-D space as follows:

$$y = \sum_{i=1}^{n} \Phi_\mathbf{i}\psi_i. \tag{7}$$

A. Problem Formulation

GE framework can be employed for finding an optimal mapping function, projecting high-dimensional data into a low-dimensional space by minimizing the following objective function:

$$\min_y \sum_{ij} (y_i - y_j)^2 \mathbf{W}_{ij} \tag{8}$$

where y_i refers to the coordinate of ith point in the low-dimensional space and $\mathbf{W}_{ij}$ is a weight between the ith and jth samples in the neighborhood of the k-nearest in ambient space. The objective function can be rewritten in the following matrix form:

$$\begin{aligned}&\min_y \quad \mathbf{y}^T\mathbf{L}\mathbf{y}\\ &\text{s.t.} \quad \mathbf{y}^T\mathbf{D}\mathbf{y} = 1\end{aligned} \tag{9}$$

where $\mathbf{L}$ is the graph Laplacian matrix, calculated as $\mathbf{L} = \mathbf{W}-\mathbf{D}$, where $\mathbf{W}$ is an $n \times n$ adjacency matrix whose elements are $\mathbf{W}_{ij}$, and the $\mathbf{D}$ is a diagonal degree matrix of $\mathbf{W}$ such that $\mathbf{D}_{ii} = \sum_j \mathbf{W}_{ij}$.

Let us set the $r = 1$, so that $y \in \Re^{n\times 1}$ is a 1-D vector. The first-order HDMR expansion given in (7) is plugged into the objective function (9), leading to

$$y = \sum_{i=1}^{d} \psi_i^T \Phi_\mathbf{i}{}^T \mathbf{L} \sum_{j=1}^{d} \Phi_\mathbf{j}\psi_j. \tag{10}$$

We hereby define the orthogonality condition of the HDMR terms using Dirac delta function δ_{ij} as the following discrete representation:

$$\psi_i^T \Phi_\mathbf{i}{}^T \Phi_\mathbf{j}\psi_j = \delta_{ij} \tag{11}$$

where δ_{ij} equals to 0 when $i \neq j$ and 1 otherwise. If this orthogonality condition is imposed as a constraint to the objective function given in (9), the optimization problem of the proposed method becomes

$$\begin{aligned}&\underset{\psi}{\text{minimize}} \quad \sum_{i=1}^{n} \psi_i^T \Phi_\mathbf{i}{}^T \mathbf{L} \sum_{j=1}^{n} \Phi_\mathbf{j}\psi_j\\ &\text{s. t.} \quad \psi_i^T \Phi_\mathbf{i}{}^T \Phi_\mathbf{j}\psi_j = \delta_{ij}.\end{aligned} \tag{12}$$

An unconstrained optimization problem can be obtained by including Lagrange multipliers and the related orthogonality conditions ($\psi_k^T \Phi_\mathbf{k}{}^T \Phi_\mathbf{i}\psi_i = 0$ for $i \neq k$ and $\psi_k^T \Phi_\mathbf{k}{}^T \Phi_\mathbf{k}\psi_k = 1$)

$$2\psi_k^T \Phi_\mathbf{k}{}^T \mathbf{L}\Phi_\mathbf{k}\psi_k + \sum_{\substack{i=1\\ i\neq k}}^{n} \psi_k^T \Phi_\mathbf{k}{}^T \mathbf{L}\Phi_\mathbf{i}\psi_i = 2\lambda_k. \tag{13}$$

Lemma 1.1: Given $\Phi_\mathbf{i}$, ψ_i and $\mathbf{L}$ whose dimensions are $n \times p$, $p \times 1$ and $n \times n$, respectively, the equality $\mathbf{L}\Phi_\mathbf{i}\psi_i = \lambda\Phi_\mathbf{i}\psi_i$ holds if λ is one of the eigenvalues of below equation

$$\Phi_\mathbf{i}{}^T \mathbf{L}\Phi_\mathbf{i}\psi_i = \lambda\Phi_\mathbf{i}{}^T \Phi_\mathbf{i}\psi_i. \tag{14}$$

Proof: We multiply (14) with $\Phi_\mathbf{i}$ and then $(\Phi_\mathbf{i}\Phi_\mathbf{i}^\mathbf{T})^\dagger$ from left, yielding to

$$\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T \mathbf{L}\Phi_\mathbf{i}\psi_i = \lambda\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T \Phi_\mathbf{i}\psi_i,$$
$$\underbrace{\left(\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T\right)^\dagger \Phi_\mathbf{i}\Phi_\mathbf{i}{}^T}_{\approx I} \mathbf{L}\Phi_\mathbf{i}\psi_i = \lambda \underbrace{\left(\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T\right)^\dagger \Phi_\mathbf{i}\Phi_\mathbf{i}{}^T}_{\approx I} \Phi_\mathbf{i}\psi_i.$$

Note that, $\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T$ is an $(n \times n)$ matrix with r-rank. Therefore, one can replace $\mathbf{L}\Phi_\mathbf{i}\psi_i$ with $\lambda\Phi_\mathbf{i}\psi_i$ in the least-squares sense. Here, $(\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T)^\dagger$ corresponds to the generalized Moore–Penrose inverse and can be easily calculated by using the singular value decomposition of $\Phi_\mathbf{i}\Phi_\mathbf{i}{}^T$.

According to Lemma 1.1, we can therefore replace the matrix $\mathbf{L}$ with one of its eigenvalues given in (13), yielding to

$$2\psi_k^T \Phi_\mathbf{k}{}^T \mathbf{L}\Phi_\mathbf{k}\psi_k + \sum_{\substack{i=1\\ i\neq k}}^{n} \psi_k^T \Phi_\mathbf{k}{}^T \lambda\Phi_\mathbf{i}\psi_i = 2\lambda_k$$
$$2\psi_k^T \Phi_\mathbf{k}{}^T \mathbf{L}\Phi_\mathbf{k}\psi_k + \sum_{\substack{i=1\\ i\neq k}}^{n} \lambda \underbrace{\psi_k^T \Phi_\mathbf{k}{}^T \Phi_\mathbf{i}\psi_i}_{=0} = 2\lambda_k$$
$$\psi_k^T \Phi_\mathbf{k}{}^T \mathbf{L}\Phi_\mathbf{k}\psi_k = \lambda_k. \tag{15}$$

Equation (15) can be rearranged in a more compact format as follows:

$$\psi_k^T \mathbf{A}_k\psi_k = \lambda_k \tag{16}$$

where $\mathbf{A}_k$ is a $p \times p$ matrix and defined as $\mathbf{A}_k = \Phi_\mathbf{k}^\mathbf{T}\mathbf{L}\Phi_\mathbf{k}$. It should be noted that $\Phi_\mathbf{k}$ is an orthogonal matrix due to the orthogonality of Legendre polynomials, and the $\mathbf{L}$ is a symmetric matrix; therefore, $\mathbf{A}_k$ is also symmetric [63]. The eigenvectors of the eigensystem in (16) are orthogonal to each

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other without losing any generality; hence we can assume that they are orthonormal as well. Moreover, based on the spectral theory discussed in [63], one can say that (16) is equivalent to the eigenproblem of $\mathbf{A}_k\psi_k = \lambda_k\psi_k$. Therefore, solving d separated p-dimensional eigenproblems instead of solving one n-dimensional problem will produce the same information.

Henceforth, it is possible to quickly calculate the related eigenvalues of each feature by solving a p-dimensional eigenproblem instead of dealing with an n-dimensional problem where $p \ll n$. The proposed method provides an effective divide-and-conquer strategy, which reduces the communication cost of the overall FS mechanism, particularly for big data analysis in modern high-performance computing systems. In the era of EO big data processing and exascale computing, the primary bottleneck of the large-scale problems that prevent the algorithm's scalability is the communication needs of the algorithms. Therefore the current trend in computational linear algebra is to redesign the well-known algorithms to reduce their communication needs, even sometimes with the cost of increased computational needs [64], [65]. Since there is no algorithmic connection between the decomposed eigenvalue problems in the proposed approach, it can be considered a proper candidate for large-scale FS mechanisms in exascale computing platforms. In this study, we focused on the reliability of the proposed mechanism, and in future work, this approach will be implemented in high-performance computing systems.

B. On the Connection to Sensitivity Analysis

It should be noted that using eigenvalues as feature relevancy scores raises an essential question since they are not guaranteed to be the eigenvalues of the Laplacian matrix; hence, they might seem irrelevant in the context of FS. Since we imposed an orthogonality constraint on the optimization problem, allowing a variance decomposition, we will leverage another useful property of the HDMR, namely sensitivity analysis, to connect the proposed method to the FS. The HDMR has the ability to present the effect of each variable on the model output due to the orthogonality condition. To determine the effect of each feature on the model output, the first-order feature relevance score associated with each feature can be calculated using the coefficients of the LPs as follows:

$$\psi_i = \frac{D_i}{D}, \quad i = 1, \ldots, d. \tag{17}$$

The individual variance D_i and total variance D are calculated as follows:

$$D_i = \sum_{j=1}^{p} (\tau_j)^2, \quad D = \sum_{i=1}^{d} D_i. \tag{18}$$

As discussed in Section I, the primary purpose of this study is to find proper and easy-to-compute feature relevance factors to measure the distinct effect of each feature on the nonlinear mapping function $f(\mathbf{X})$. The algorithm of the proposed method is presented in Algorithm 1, and the code is provided in the GitHub repository at https://github.com/IPL-UV/HDMR4FS.

Algorithm 1 Orthogonal HDMR-Based FS

Input(s): Data matrix: $\mathbf{X}^{n\times d}$, Order of Legendre polynomials: p

Output(s): Feature relevance score: τ_i for each feature

```
Generate augmented data matrix Φ with Legendre's.
Calculate weight matrix W from augmented data matrix Φ
Calculate graph Laplacian L = W − D
for i ← 1 to n do
    Form A_i = Φ_i^T L Φ_i
    Form0 B_i = Φ_i^T Φ_i
    Solve eigenproblem of A_i ψ_i = λ_i B_i ψ_i
end for
Normalize each ψ_i vector,
Calculate feature relevance score τ_i = ||ψ_i||^2
Sort τ_i values in descending order.
```

C. On the Computational Complexity (CC)

The CC analysis of the proposed method can be given as follows.

1) Construction of extended data matrix with Legendre polynomials $\mathcal{O}(n^2dp)$ where p represents the degree of Legendre polynomials.
2) Construction of Laplacian matrix $\mathcal{O}(n^2d)$.
3) Construction of p-dimensional eigenproblems via sparse matrix–vector (SpMV) operations $\mathcal{O}(pnk)$ where k is the maximum number of non-zeros in matrix $\mathbf{L}$.
4) p-dimensional eigenproblem solution $\mathcal{O}(dp^3)$ for d times. Note that $\mathcal{O}(p^3)$ is considered as the worst case scenario, which is negligible as $p \ll n$.

Therefore, overall complexity of the proposed approach is $\mathcal{O}(n^2d(1+p)+pnk+dp^3)$. Since $k \ll n$, $p \ll n$ and in general $d < n$, the CC of the proposed method will be dominated by the n^2, leading to $\mathcal{O}(n^2)$. Since the analysis shown here assumes that the matrices are dense, it can be regarded as the worst case scenario. However, the graph matrices are usually sparse; therefore, the eigenvalue decomposition's complexity for the sparse matrices is far lesser than the given complexities.

V. Experimental Setup

Let us briefly describe the data collection and experimental setup in the experiments.

A. Hyperspectral Image Datasets

We perform comprehensive experiments on three HSI, with the following descriptions.

1) Kennedy Space Center (KSC): The Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor of the National Aeronautics and Space Administration (NASA) acquired airborne hyperspectral data with an 18-m spatial resolution above KSC in March 1996. Noisy and water absorption bands were deleted, leaving 176 features for 13 different wetland and upland classes. In this dataset, the image has a spatial size of 512 × 614 different land cover classes. Due to the similarity of spectral signatures across certain vegetation types, distinguishing distinct land covers is challenging.

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2) Botswana: The Botswana hyperspectral data comprise of Hyperion image taken over Botswana's Okavango Delta in May 2001 with nine classes of complex natural vegetation. The general class groupings include seasonal swamps, occasional swamps, and woodlands. The signatures of several classes are spectrally overlapped, typically leading to low classification accuracies. There are 145 bands after water absorption; noisy and overlapping spectral bands are eliminated. The image has a 30-m spatial resolution and a wavelength range of 400–2500 nm.

3) Indian Pines: The Indian Pines image, acquired by the Airborne Visible/Infrared Imaging Spectrometer above the agricultural Indian Pine test site in Northwestern Indiana (AVIRIS), includes 220 bands spanning from 400 to 2500 nm across the electromagnetic spectrum, with spatial coverage of 45 × 145 pixels and a spatial resolution of 20-m/pixel. The image has 16 ground-truth classes with 200 spectral bands after removing 20 water absorption bands.

B. Benchmark Methods

We implemented and compared nine standard FS approaches, most of which are graph-based. This section provides a brief overview of the approaches.

1) Laplacian Score [38]: It is an unsupervised method, assigning a relevancy score to each feature through the Laplacian matrix. The LapScore is essentially based on Laplacian Eigenmaps [39], including a generalized eigenvalue problem as $\mathbf{L}y = \lambda \mathbf{D}y$.

2) Unsupervised Discriminative FS [51]: The UDFS is an unsupervised method and is based on the simultaneous exploitation of local discriminative information and feature correlations. Despite its unsupervised nature, the UDFS assumes that class labels can be predicted by a linear classifier; therefore, the method calculates a local scatter matrix and a between-class scatter matrix to define a local discriminative score. This score (the larger, the better) can be obtained by minimizing the sparse regularization term by $\ell_{2,1}$-norm.

3) FS With Adaptive Structure Learning (FSASL) [66]: The method adaptively uses both local and global structure learning by solving a complex optimization problem that results in a higher computational cost than the previous methods.

4) Local Learning-Based Clustering FS (LLCFS) [67]: The LLCFS follows the same approach as the FSASL. As stated in [66], the main difference between these arises from the unified objective function for structure learning and the feature search. However, their computational complexities are equivalent since both approaches have similar computational procedures.

5) Dependence Guided UFS (DGUFS) [68]: The DGUFS is a graph-based sparse learning approach aiming to select features and perform clustering jointly.

6) FS Concave (FSV) [69]: The primary purpose of this method is to select the subset of features by employing the nonlinear support vector machine (SVM) that is used for the solution of the related optimization problem.

7) Recursive Feature Elimination (RFE) [70]: This algorithm is based on applying the SVM algorithm to find the weights of each feature vector, where the features are ranked with an RFE strategy.

8) Multicluster FS [50]: The features, preserving the multicluster nature of the data, are selected by applying spectral analysis techniques. The main computational kernels of the algorithm are the sparse eigenvalue problem and an ℓ_1-norm regularized least-squares problem.

9) Spectral FS (SPEC) [40]: The method is based on a graph-theoretical approach and forms feature ranking functions according to the effect of the features on the structures of the graph induced from the incidence similarity matrix.

C. Experimental Design

The expressiveness of the selected features is evaluated in terms of classification accuracy with methods such as k nearest neighbors (k-NNs), SVMs with RBF kernel, and linear discriminant analysis (LDA). This selection is not incidental: we aim to assess the level of nonlinearity captured by the FS methods when selecting features. The parameters of all the classifiers are tuned based on fivefold cross validation.

In graph-based FS methods used in the comparisons, the construction of the graph Laplacian matrix is the most significant parameter affecting the performance of the FS method. Therefore, to reduce the effect of the graph matrix on the performance of the FS methods, the weights of connected samples are constructed based on the binary relations, meaning that they are set as 1 if vertices i and j are connected by an edge and are 0 otherwise. Note that all the samples are connected in an unsupervised fashion without using prior class label information when constructing the graph Laplacian matrix. In order to ensure a fair performance comparison between graph-based FS methods, the same graph adjacency matrix is utilized across all implementations. As for the proposed method, another parameter, the order of Legendre polynomial, is empirically found as three. This selection has been consistently observed to result in superior performance on validation sets across all hyperspectral images (HSI).

To evaluate the performance of FS methods, metrics such as overall accuracy (OA), class-wise accuracy, Kappa coefficient, and McNemar's test are used. The OA measures the proportion of the correctly sampled sample in all test samples, while the Kappa coefficient measures the accuracy of classification by comparing the observed agreement with the expected agreement due to chance alone. Additionally, McNemar's test is a widely used metric for statistically comparing the performance of two algorithms and is computed as follows:

$$Z\text{-score} = \frac{g_{12} - g_{21}}{\sqrt{g_{12} - g_{21}}} \tag{19}$$

where g_{12} corresponds to number of samples correctly classified by classifier-1 but incorrectly classified by classifier-2, while g_{12} is exactly the opposite.

The learning curves for each HSI dataset are obtained as a function of average classification accuracy and selected features over ten trials. In all the experiments, the training and test are randomly generated from the ground-truth HSI dataset, with a ratio of 10% for the training set and 90% for the test set. We assess the statistical significance of the FS methods using

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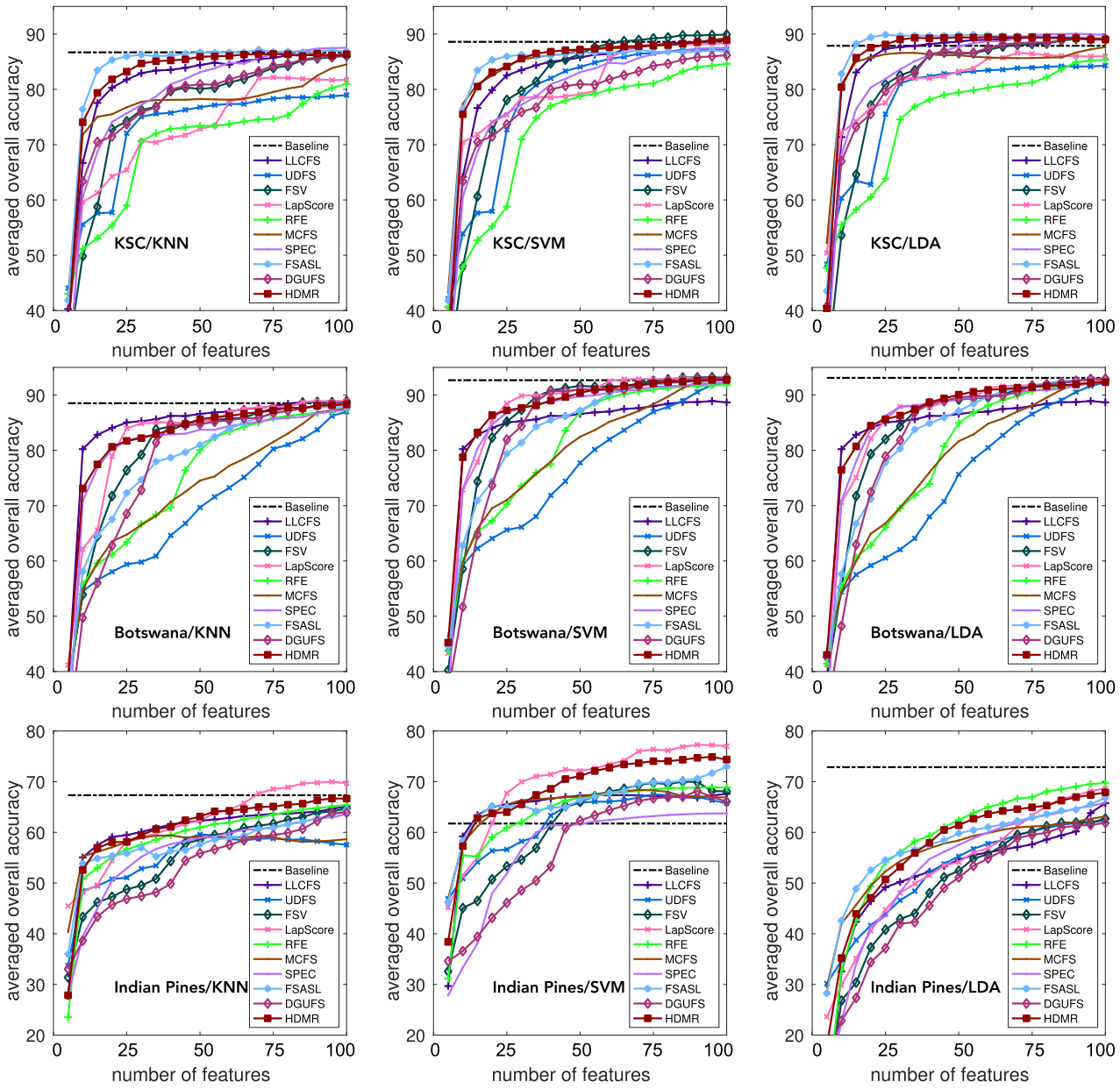


Fig. 1. Average overall classification accuracy for each HSI over ten trials versus the number of selected bands provided by each FS method.

McNemar's test, which yields the Z-scores of the methods at the first ten, 25, and 50 selected features. Performance is also evaluated in terms of per-class accuracy and visually through thematic maps. Finally, the histogram of the selected features is illustrated to observe how frequently they are selected over ten runs.

VI. Experimental Results

The performance of the proposed method is compared to that of competitive methods for each HS in terms of averaged OA achieved over ten random training and test splittings. Fig. 1 shows the learning curves of accuracy achieved at the selected features by each FS method using KNN, SVM, and LDA classifiers.

According to the results in Fig. 1, the HDMR performs competitively in each HSI with a different classifier compared to the state-of-the-art approaches, demonstrating that the selected features by the HMDR are mostly discriminative for each classification method. The proposed method is typically followed by the FSASL, LLCFS, and spectral FS (SPEC), which all have a high classification accuracy. For example, the FSASL performs slightly better than the HDMR in the KSC dataset when the number of selected features is between 5 and 50; it does not perform well in Botswana, where the HDMR outperforms it significantly. Similarly, the FSASL's performance with KNN and SVM is significantly worse than the HDMR in Indian Pines dataset, demonstrating that the FSSL is not superior to the HDMR. The LapScore outperforms the HDMR in Indian Pines/SVM, especially after the first 25 features are selected, whereas the HDMR generally outperforms LapScore with KNN and LDA classifiers. Almost in all HSI datasets, the LLCFS is competitive with the HDMR. However, it does

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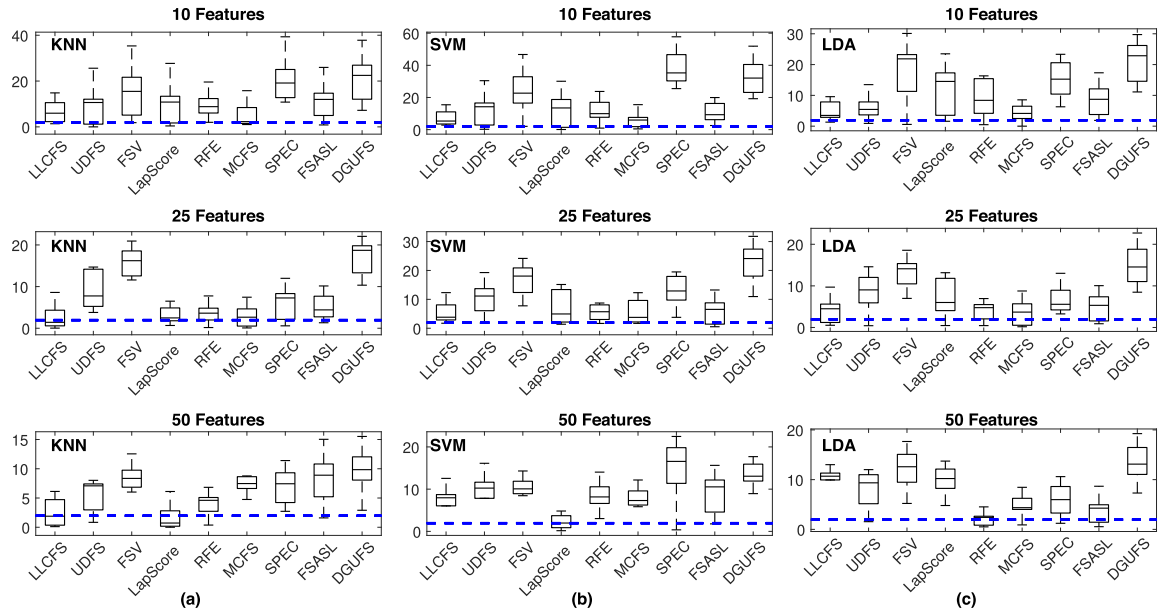


Fig. 2. Assessing the sensitivity of classification accuracy in terms of the Z-statistic of HDMR versus other FS methods for Indian Pines dataset with (a) KNN, (b) SVM, and (c) LDA classification.

not outperform the HDMR, especially in the first 25 selected features of the KSC dataset, Bostwana/SVM, Bostwana/LDA, Indian Pines/SVM, and Indian Pines LDA. The SPEC, on the other hand, performs well in the Botswana dataset but fails in the KSC and Indian Pines datasets. Although the remaining FS methods perform well in certain cases, they do not behave consistently across all experiment configurations, whereas the proposed method does.

The Z-statistic is used to compare the average OAs for the first ten, 25, and 50 selected features only in the Indian Pines dataset to see if the associated FS technique is significantly different from the others. The box plots of Z-scores over ten trials are shown in Fig. 2 for HDMR versus other methods, with a blue line showing 1.96 of Z-score, suggesting that two results are statistically different at the 95% confidence level. Although LLCFS, LapScore, and RFE perform similar to the HDMR when using KNN in the Indian Pines dataset, the HDMR is statistically different from all of the other FS methods, as shown in Fig. 2(a), except for the Laplacian method, which yields Z-score of 1.67 on average only when 50 features are used. As shown from Fig. 2(b), average Z-scores of Laplacian are greater than 1.96, indicating that the HDMR is statistically different than Laplacian at the first ten, 15, and 50 features, but Laplacian generally performs better than the HDMR, especially when considering the features between 25-50 and 75-100. However, as seen in Fig. 2(c), it should be noted that Laplacian achieves relatively low classification accuracy, resulting in much higher Z-scores than HDMR when employing LDA. The RFE and FSASL perform well in LDA classification, but they do not perform well in other classification settings. Based on these findings, we may conclude that the selected features by the Laplacian are not more discriminative than HDMR's features across all the classifiers despite Laplacian's classification accuracy being comparable to the HDMR. Since the results were similar, we did not include the other HSI dataset in the analysis.

Indian Pines dataset is classified with SVM, mainly utilizing the top 50 features selected by the associated FS method to observe the selected features' effectiveness visually. Fig. 3 shows the thematic maps, OA, and Kappa values. Because the number of FS methods utilized in the comparisons is relatively high, we only provided thematic maps for a few representative FS methods that performed well and poorly on the test data, including the scenario of using all features (Baseline). Regarding overall classification accuracy and Kappa value, HDMR performs similar to the Laplacian but is much better than all the other FS methods. We also present the class-wise accuracy to quantitatively compare the performance of the selected features per class, as shown in Table I.

As can be seen in Table I, the HDMR outperforms the other FS methods, especially for the classes of *grass pasture* (5), *grass trees* (6), *soybeans-min till* (11), and *hay-windrowed* (15), which can also be visually observed from the thematic maps provided in Fig. 3. Although the FSASL, MCFS, and RFE achieve the highest classification performances for classes of *corn-no till* (2), *corn* (8), and *soybeans-no till* (10), respectively, the HDMR performs the second-highest performance on these classes even with very close performances. It should be noted that very low classification performances are achieved for classes of *alfalfa* (1), *grass-pasture-mowed* (7), and *oats* (9), which might be due to the limited size of the training set.

The FS methods are also analyzed to determine which features are selected and how sensitive they are to training/test splits independently of any classifier. Fig. 4 shows histogram plots of all the top 25 features of the associated FS method selected on ten different random realizations of the training/test

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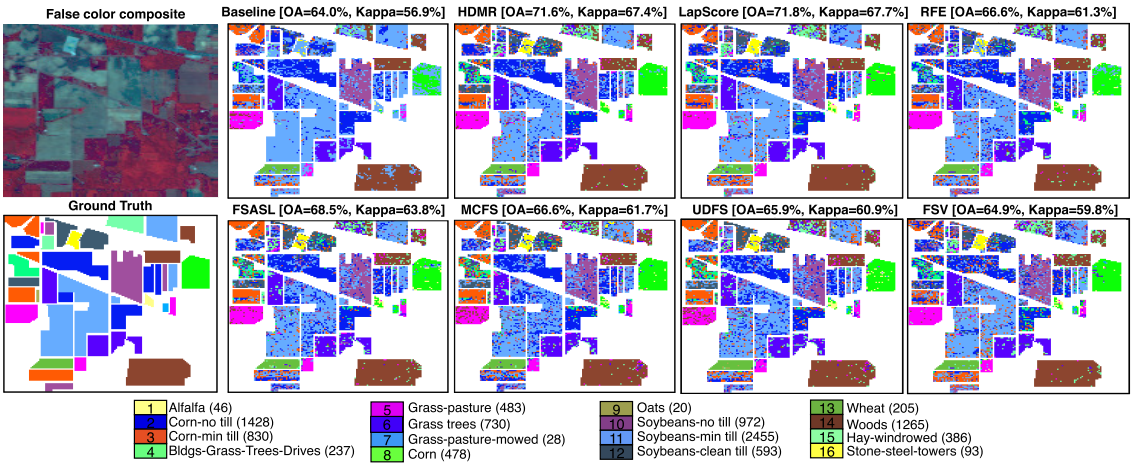


Fig. 3. Thematic maps of the Indian Pines dataset obtained with SVM utilizing the top 50 features selected by the associated FS method.

TABLE I

CLASS-WISE ACCURACY OF THE TEST DATA USED IN GENERATING THEMATIC MAPS OF INDIAN PINES DATASET WHEN USING SVM CLASSIFIER. BOLD AND UNDERLINED FONTS REPRESENT THE FIRST-HIGHEST AND THE SECOND-HIGHEST CLASSIFICATION ACCURACIES

Classes	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
Training	5	143	83	24	49	73	3	48	2	98	246	60	21	127	39	10
Test	41	1285	747	213	434	657	25	430	18	874	2209	533	184	1138	347	83
Baseline	4.9	62.5	46.1	5.2	62.9	79.6	0.0	48.6	0.0	54.5	**90.2**	15.2	77.2	88	11.2	6
HDMR	17.1	65.8	53.4	36.2	**85.7**	**91.5**	16.0	93.6	5.8	59.4	77.4	50.8	89.1	91.8	**38.6**	68.7
LLCFS	7.3	54.5	50.2	23.5	80.0	87.4	44.0	80.7	0.0	58.6	73.2	31.7	90.8	88.6	38.3	65.1
UDFS	56.1	58.3	**59.4**	33.8	68.9	77.9	24	89.8	5.6	48.5	68.6	**58.2**	**94.0**	88.2	26.5	78.3
FSV	41.5	51.6	44.8	35.2	83.2	88.9	20	93.5	5.6	50.0	69.5	32.6	89.7	89.9	38	81.9
LapScore	**63.4**	64.3	54.9	**41.3**	85.3	88.7	**60**	93	**16.7**	57.3	78.9	45.8	84.2	**93.4**	35.7	**84.3**
RFE	24.4	58.7	48.6	19.7	80.6	81.3	32	80.2	0.0	**60.0**	77.2	30.6	91.8	92.2	26.8	36.1
MCFS	31.7	63.1	59.0	26.8	70.3	78.8	16	**94.2**	5.6	50.9	67.5	57.6	89.1	88.9	31.7	72.3
SPEC	2.4	52.4	37.6	4.2	74.0	89.2	4.0	59.8	0.0	44.7	76.7	19.5	89.1	90.3	21.3	19.3
FSASL	36.6	**66.5**	56.9	30.0	80.0	81.1	16	87.7	5.6	51.7	73	55.5	88.0	88.4	32	66.3
DGUFS	26.8	47.4	36.8	35.2	79.0	90.9	20	92.8	0.0	47.7	67.8	30.4	89.1	88.4	34	77.1

splits, along with the spectral signatures of each class of interest. As shown, there is no consensus across the FS approaches regarding where the features should be selected, meaning that the FS problem has no unique solution. Some FS methods find the essential features from the visible spectral range, while others look for them in the red edge, both near-infrared (NIR) and/or short-wave infrared bands. This makes sense, as many crop and vegetation classes are present in the images. For example, in the Bostwana dataset, the FS methods, including FS, LapScore, and DGUFS, select the features from the visible part of the spectrum. In contrast, some, UDFS, MCFS, and DGUFS, mainly select the features from SIR region. The features selected by the HDMR are generally evenly distributed across the wavelength, covering all of the spectrum, which might be why the HDMR performs well in all the datasets and the classifiers. On the other hand, the ideal FS method may be the one that provides the highest classification accuracy and the same set of chosen features independent of the different training/test split realizations. In this regard, the HDMR achieves the highest classification accuracy with its selected features. Still, it lacks the stability of the selected features compared to the ones, including UDFS, LapScore, RFE, SPEC, and DGUFS.

We conducted further analysis to demonstrate the distribution of the selected features for the proposed method across the electromagnetic spectrum by varying the rate of the training sample on the KSC dataset. Fig. 5 depicts the frequency distribution of the first 25 selected features obtained from the proposed method, using 20 different random realizations of training-test splitting. Each subplot in Fig. 5 corresponds to a different dataset rate used as the training set. The figure illustrates that, when using a 10% training set, the selected features cover most of the spectrum. However, as the training set rate increases, the features become less selected in the visible region. Despite this, the overall distribution of the selected features across the electromagnetic spectrum remains relatively stable, suggesting that the proposed method may be robust in FS and unaffected by variations in training set size.

In Table II, we compare the computational complexities of the proposed method and its competitors. Note that n, d, r represent the number of samples, dimensionality, and selected features. Several algorithm-specific parameters in Table II, such as p, k for HDMR, represent the required number of Legendre polynomials and the number of non-zeros in the Laplacian, respectively. As shown in Table II, the complexity of the proposed approach is in $[\mathcal{O}(n^2)]$, similar to SPEC, MCFS, and RFE and lower than the other methods. However, the main advantage of the proposed approach arises from its algorithmic setup based on the divide-conquer methodology. This will strengthen the approach's applicability for large-scale

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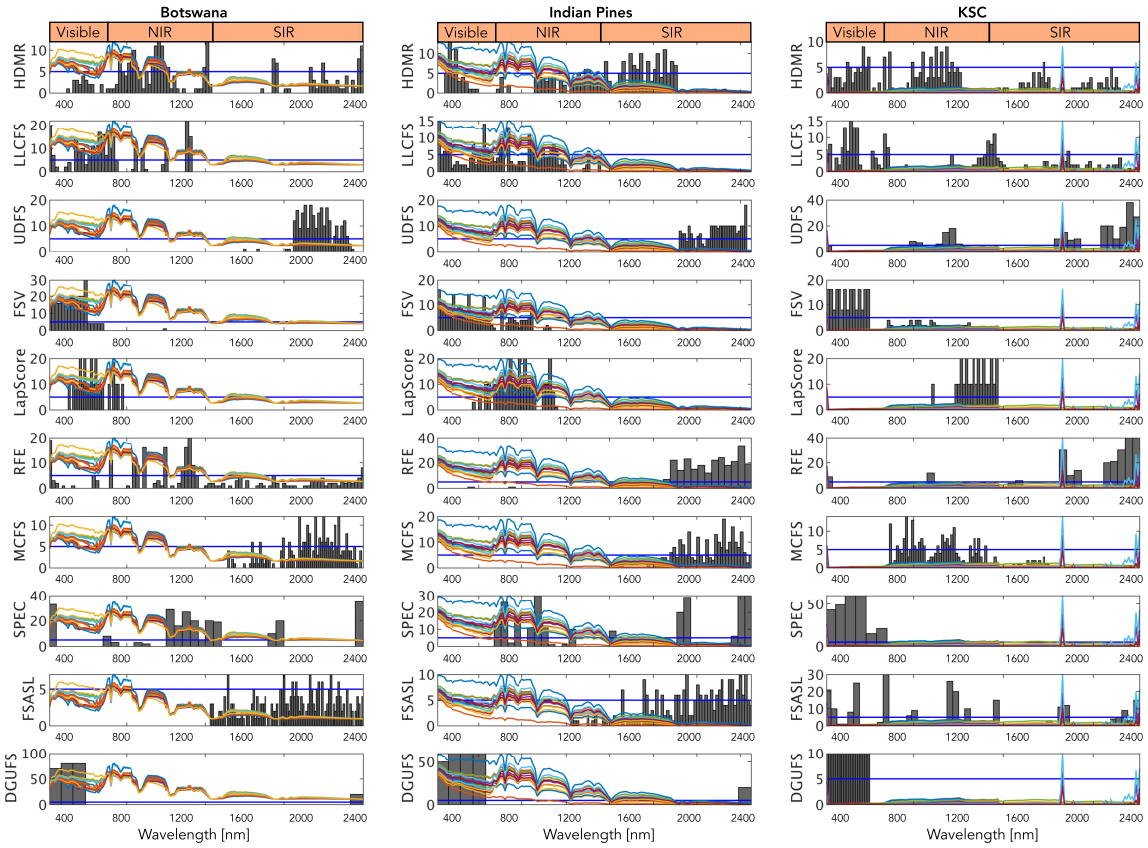


Fig. 4. Histogram plot of the first 25 selected features for each FS method over ten different random realizations. The blue horizontal line shows the frequency of selected features above 5.

TABLE II
COMPARISON OF THE FS METHODS

FS Method	Type	Category	Methodology	Computational Complexity
HDMR	unsupervised	filter	manifold + similarity	$\mathcal{O}(n^2d(1+p)+pnk+dp^3)$
LapScore	unsupervised	filter	manifold + similarity	$\mathcal{O}(d^2+n^2)$
SPEC	unsupervised	filter	similarity	$\mathcal{O}(dn^2)$
LLCFS	unsupervised	embedded	structural learning + clustering	$\mathcal{O}(n^3+n^2)$
UDFS	unsupervised	embedded	sparse learning	$\mathcal{O}(d^2+n^3)$
MCFS	unsupervised	embedded	sparse learning	$\mathcal{O}(dn^2)$
FSASL	unsupervised	embedded	structural learning + clustering	$\mathcal{O}(n^3+n^2)$
DGUFS	unsupervised	wrapper	clustering with $\ell_{2,0}$-norm	$\mathcal{O}(n^3+n^2+nd)$
FSV	supervised	wrapper	linear programming + SVM	$\mathcal{O}(n^3)$
RFE	supervised	wrapper	sequential feature elimination + SVM	$\mathcal{O}(n^2dlog_2d)$

datasets on parallel architectures. It should be noted that the most significant contribution of our method is to reduce the complexity of the eigenvalue problem from $\mathcal{O}(n^3)$ to $\mathcal{O}(p^3)$ in the worst case scenario. However, if one employs advanced eigensolvers to exploit the sparsity of the eigenproblem arising from (9), the proposed approach will remain less costly since $p \ll n$.

Another analysis carried out in this study is to measure the computational time of the FS methods while selecting features. Table III lists the times in seconds for each FS method and dataset. As can be observed, the proposed method takes much less time than others, such as RFE, LLCFS, MCFS, and FSASL. It should be noted that computational times reported in this article may vary based on the computer used and the coding strategies used when implementing the FS methods, such as whether or not a serial code optimization is performed; therefore, the results may be biased. Therefore, the CC of the methods presents more reliable information than

TABLE III
TIMES IN SECOND WHEN SELECTING THE FEATURES FOR EACH HSI DATASET

Methods	KSC	Indian Pines	Botswana
HDMR	0.97	1.2	0.38
LLCFS	2.37	10.02	3.39
UDFS	0.47	1.45	0.55
FSV	0.62	4.72	2.49
Laplacian	0.25	**0.82**	**0.31**
RFE	11.04	1186.1	23.19
MCFS	1.74	4.89	1.5
SPEC	**0.22**	1.2	0.49
FSASL	2.66	811.6	201.58
DGUFS	0.84	3.82	1.22

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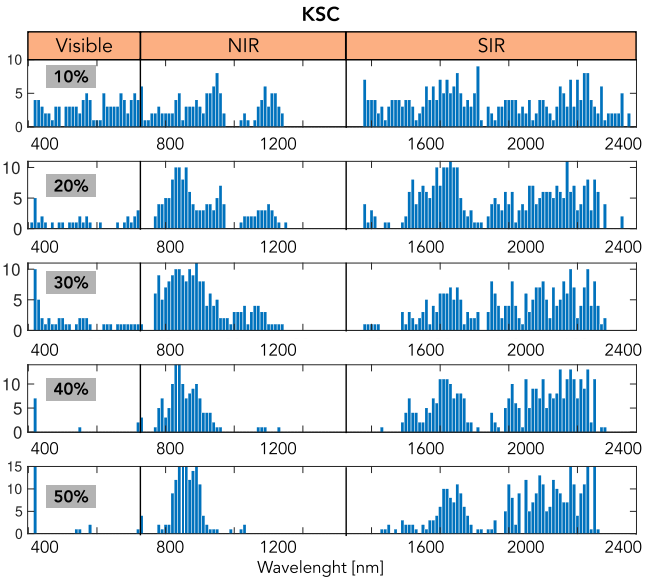


Fig. 5. Illustration of the first 25 selected features across the electromagnetic spectrum with varying rates of training samples.

the CPU-time elapsed when evaluating the efficiency of the methods.

VII. Conclusion

GE is a very efficient method to reduce the dimensionality, but solving a generalized eigenvalue problem becomes a computationally burdensome process, especially for large-scale and high-dimensional problems. We propose a graph-based UFS method using the orthogonality condition of the method, called HDMR. The solution of the optimization problem leads to a set of small-sized eigenvalue problems. Thanks to the global sensitivity of the HDMR, the proposed method is used as an FS method. The most time-consuming part of the proposed algorithm is the construction of the extended data matrix and the Laplacian. However, these steps have independent computations and can be handled in parallel for large-scale data. Our method has the following advantages.

1) Due to its embarrassingly parallel algorithmic setup, the proposed method is scalable for high-dimensional problems.
2) As it surveys the structural neighborhood of data using GE, the locality-preserving properties of the selected features are attained.
3) It handles feature relevancy with a new graph-based approach incorporating global sensitivity analysis.
4) The data is explicitly mapped into a high-dimensional space using Legendre polynomials where nonlinearity is better to be modeled.
5) One of the significant drawbacks of GE is that the extracted features are not interpretable. Our proposed method's extracted features are interpretable as the data is explicitly mapped to a high-dimensional space using Legendre polynomials.
6) It is an unsupervised method that does not require any labeled data.

On the other hand, the most crucial advantage of the methodology will be more visible for big EO data applications, which is the biggest challenge of modern FS approaches. The future direction of this study will be to utilize the graph FS schemes in deep learning models to improve the explainability of the results.

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